Software Engineering

Lab 3: Component Modelling & Architectural Pattern Selection

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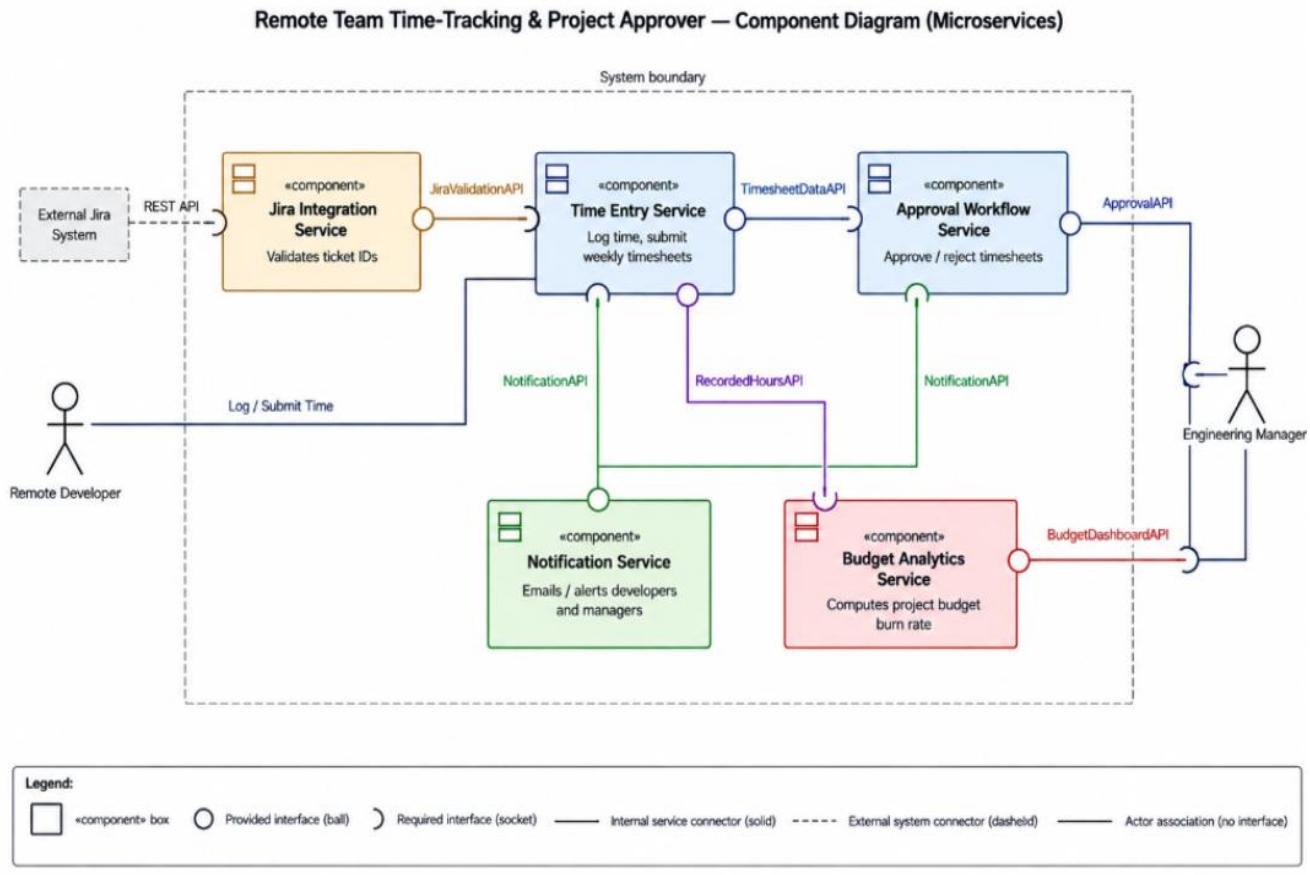
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SECTION: A

SEM: 5

Problem Statement - Remote Team Time-Tracking & Project Approver

Component digaram



Architecture Selection & Justification

I chose Microservices Architecture for the Remote Team Time-Tracking & Project Approver System.

Architectural Choice: Microservices Architecture, with five independently deployable services: Time Entry Service, Approval Workflow Service, Budget Analytics Service, Jira Integration Service, and Notification Service. These services communicate through clear interfaces such as JiraValidationAPI, TimesheetDataAPI, RecordedHoursAPI, NotificationAPI, ApprovalAPI, and BudgetDashboardAPI.

Two Reasons

1. Natural alignment with system boundaries: The system's requirements are divided into independent capabilities in the Lab 2 backlog: Time Logging & Submission, Approval Workflow, and Budget Visibility. These capabilities can be implemented as independently deployable services with clear interfaces, allowing changes to approval logic without redeploying the time-entry or budget-analytics services.

2. Isolation of the external Jira dependency: The Jira Integration Service handles ticket-ID validation through the external Jira system, as specified in the Lab 2 backlog through the "Validate Jira Ticket on Entry" user story. Separating this dependency prevents Jira-related delays or failures from directly affecting the core time-logging and timesheet-submission services.

Security Advantage

Each service can enforce its own authentication and authorization rules rather than relying on one shared access-control layer. For example, only authorized Engineering Managers should be permitted to approve or reject timesheets, in accordance with NFR-002. If one service is compromised, the impact can be limited to that service's own data and responsibilities.

Performance Benefit

The Budget Analytics Service, which must refresh project budget burn information within 500 ms in accordance with NFR-001, can be scaled and cached independently of the higher-volume Time Entry Service. During peak submission periods, time-entry traffic can increase without unnecessarily slowing down budget-dashboard calculations.